PES UNIVERSITY

Department of Computer Science and Engineering

SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT

Software Test Plan

Electronics Store Inventory Management System (ESIMS)

Project	Electronics Store Inventory Management System (ESIMS)
Group / Project ID	Team 8
Version	1.0
Date	30-09-2026
Technology	Python (Django), PostgreSQL
Methodology	Agile

Revision History

Version	Date	Author	Change Summary
1.0	30-09-2026	Team 8	Completed functional, non-functional, security and traceability planning.

1. Introduction

Purpose: This Software Test Plan defines the testing objectives, scope, strategy, environment, test cases, security validation, traceability and reporting approach for ESIMS Version 1.0.

Scope: Testing covers authentication/RBAC, product and category management, stock-in/out, supplier management, low-stock alerts, reporting, search/filtering and the stated non-functional and security requirements. Out-of-scope items from the SRS, such as barcode hardware, POS hardware, payment gateways, ERP integration and AI demand forecasting, are not tested.

References: ESIMS SRS v1.0; ESIMS SAD v1.0; PES University Software Engineering Level 3 Student Submission Guide; Mini-Project Deliverables Part-1.

Definitions: FR – Functional Requirement; NFR – Non-Functional Requirement; SEC – Security Requirement; RTM – Requirements Traceability Matrix; UAT – User Acceptance Testing.

2. Test Items

- Authentication and session management module.
- Role-based authorization and staff-account management.
- Product and category catalog module.
- Search/filter and current-stock display.
- Stock-in and stock-out transaction module.
- Supplier management module.
- Low-stock alert service.
- Inventory and sales reporting.
- Cross-cutting security, reliability, performance and usability behavior.

3. Features to be Tested

Requirement group	Feature	Primary validation
FR-001–FR-004	Login, RBAC, staff accounts, session expiry	Functional + authorization + security
FR-005–FR-009	Product/category management, search, current stock	Functional + system + performance
FR-010–FR-014	Stock-in/out, over-sale rejection, IDs, audit trail	Functional + integration + reliability
FR-015–FR-019	Suppliers, low-stock alerts, reports, top sellers	Functional + integration + system
NFR-001	95% catalog searches within 2 seconds	Performance/load test
NFR-002	99% business-hour availability	Operational review
NFR-003	Stock transaction within four screens/steps	Usability walkthrough
NFR-004	No duplicate transaction on double submission	Negative/integration test
NFR-005–NFR-006	Modularity and independent testability	Code review + unit/integration
SEC-001–SEC-006	Credential, authorization, validation, data exposure and logging controls	Security validation

4. Features Not to be Tested

- Barcode / QR hardware scanner integration.
- Point-of-sale hardware integration.
- Online customer-facing storefront.
- Payment gateway integration.
- Accounting / ERP integration.

- AI-based demand forecasting.
- Multi-store / warehouse transfer management.
- Multi-currency / multi-branch enterprise deployment.

5. Test Approach / Strategy

5.1 Test Levels

- Unit testing: service-level calculations, validation and alert generation independent of the UI.
- Integration testing: Django modules with PostgreSQL and cross-module transaction behavior.
- System testing: complete workflows through the web interface.
- Acceptance testing: representative Staff and Administrator tasks against SRS acceptance criteria.

5.2 Test Types

- Functional and negative testing.
- Regression testing after fixes or module changes.
- Performance/load testing for catalog search.
- Usability testing for stock transaction workflows.
- Reliability testing for duplicate submissions and transaction rollback.
- Security testing for authentication, RBAC, input validation, session expiry, sensitive-data exposure and security-event logging.

5.3 Entry and Exit Criteria

Entry criteria: stable test build; database schema/migrations available; representative test data prepared; test environment accessible; critical dependencies available.

Exit criteria: all planned test cases executed; all critical/high-severity blockers resolved or formally accepted; NFR thresholds evaluated; security-critical tests completed; RTM updated with execution evidence.

5.4 Test Automation Strategy

Use Django's test framework or pytest for unit/integration tests, Postman for API testing if REST is enabled, and JMeter or an equivalent load tool for the NFR-001 response-time test. CI through GitHub Actions should execute automated tests on pushes/pull requests.

5.5 Security Validation

Security objective / requirement	Validation activity	Expected evidence	Test ID
Protect credentials – SEC-001	Inspect password storage and authentication configuration.	Password hashes only; no plaintext credentials.	ST-SEC-01
Prevent unauthorized privilege – SEC-002	Attempt Staff access to Administrator-only functions.	Access denied and security event recorded.	ST-SEC-02
Validate/sanitize input – SEC-003	Submit invalid, negative, oversized and malformed values.	Request rejected without data corruption.	ST-SEC-03
Session protection – SEC-004	Wait/force inactivity timeout and reuse expired session.	Expired session requires re-authentication.	ST-SEC-04
Protect sensitive data – SEC-005	Access protected pages/API without authentication.	No sensitive supplier/pricing data disclosed.	ST-SEC-05
Security logging – SEC-006	Generate failed login and unauthorized access attempts.	Relevant events logged without passwords/secrets.	ST-SEC-06

6. Test Environment

- Server: Linux environment running Django 5.x application.
- Database: PostgreSQL 15.x.
- Clients: current Chrome, Edge and Firefox desktop browsers.
- Test tools: Django test framework / pytest, Postman if REST API is enabled, JMeter for performance, browser developer tools and GitHub Actions.

- Test data: dummy Staff/Admin accounts; active/disabled products; categories; registered suppliers; stock-in/out transactions; threshold boundary cases; duplicate submissions.
- Network: HTTPS client-server communication.

7. Test Schedule

Phase	Planned activity	Exit evidence
Test design	Derive test cases from FR/NFR/SEC requirements and SAD interfaces.	Approved test case set + RTM.
Environment preparation	Create test database, accounts, products, suppliers and baseline data.	Environment checklist.
Unit / integration	Execute service and persistence tests.	Automated test results.
System / security	Execute end-to-end workflows and security tests.	Execution log + defects.
Performance / usability	Measure search response and stock transaction steps.	Performance metrics + walkthrough notes.
Regression / UAT	Retest fixes and representative Staff/Admin workflows.	Final test summary and sign-off.

8. Test Deliverables

- Test Plan
- Test case specification
- Automated unit/integration test results
- Performance/security test evidence
- Defect reports
- Updated RTM
- Final Test Summary Report

9. Roles and Responsibilities

Role	Assigned to	Responsibility
QA / Test Coordination	Team 8 – shared	Maintain test plan, coordinate execution and consolidate results.
Feature Test Owner – Auth	Roopa Sreedhar A	Test authentication, roles and session controls.
Feature Test Owner – Catalog	Sakthevel Saravanan	Test product/category/search functionality.
Feature Test Owner – Stock	Samiksha B N	Test stock-in/out, integrity and reliability.
Feature Test Owner – Supplier/Reports	Rishitth G R	Test suppliers, alerts and reporting.

10. Risks and Mitigation

Risk	Impact	Mitigation
Unstable build	Blocks many tests and invalidates results.	Run smoke tests first; isolate blockers and retest after fixes.
Database/test-data inconsistency	False failures or incorrect stock results.	Reset controlled test fixtures before test runs.
Performance environment differs from production	Results may not represent production behavior.	Record test load/environment and compare repeated runs.
Security testing omitted from regression	Privilege/data-exposure defects may reappear.	Keep security cases in the regression suite and CI where practical.

11. Assumptions & Dependencies

- Django authentication and PostgreSQL are available in the test environment.
- Representative Staff and Administrator test accounts are available.
- The implemented application follows the SRS architecture and interfaces.
- No external payment, ERP or hardware integration is required for Version 1.0 testing.

12. Suspension & Resumption Criteria

Suspend testing when the test environment is unavailable, the build blocks a substantial portion of planned cases, or a data-integrity defect makes further execution unsafe.

Resume when the environment is stable, blocking defects are fixed or formally waived, and test data is restored to a known baseline.

13. Test Cases and Traceability

The following planned cases cover functional, non-functional and security requirements. Actual Result and Status are intentionally marked Not Executed because this document is a test plan, not an execution report.

Test ID	Req ID	Scenario	Precondition	Test Data / Action	Expected Result	Actual Result	Status
UT-AUTH-01	FR-001	Valid login	Valid Staff account	Correct credentials	Authenticated session and Staff dashboard	Not Executed	Not Executed
UT-AUTH-02	FR-001	Invalid login	Valid account	Wrong password	Login rejected; no session created	Not Executed	Not Executed
ST-AUTH-01	FR-002/ SEC-002	RBAC enforcement	Staff account	Open admin-only URL/action	Access denied; event logged	Not Executed	Not Executed
ST-AUTH-02	FR-004/ SEC-004	Session expiry	Authenticated user	Remain idle beyond configured period	Session expires; re-login required	Not Executed	Not Executed
UT-CAT-01	FR-005	Add product	Admin/ authorized user	Valid product fields	Product saved with unique SKU and stock	Not Executed	Not Executed
UT-CAT-02	FR-006	Edit product	Existing product	Change price/category	Updated details displayed	Not Executed	Not Executed
ST-CAT-01	FR-007	Disable product	Existing product with history	Disable product	Product unavailable for new sale; history retained	Not Executed	Not Executed
ST-CAT-02	FR-008	Search/filter	Catalog populated	Search by name/category/SKU	Matching products returned	Not Executed	Not Executed
ST-CAT-03	FR-009	Current stock display	Product with known quantity	Open catalog	Displayed quantity matches database	Not Executed	Not Executed
IT-STOCK-01	FR-010	Record stock-in	Valid supplier/product	Quantity > 0	Stock increases; transaction recorded	Not Executed	Not Executed
IT-STOCK-02	FR-011	Record stock-out	Active product with stock	Sell quantity <= stock	Stock decreases; transaction recorded	Not Executed	Not Executed
UT-STOCK-02	FR-012	Reject over-sale	Product stock=5	Attempt sale=6	Transaction rejected; stock remains 5	Not Executed	Not Executed
IT-STOCK-03	FR-013	Unique transaction ID	New stock transaction	Record transaction	Unique transaction ID generated	Not Executed	Not Executed
ST-STOCK-04	FR-014	Audit trail	Recorded stock transaction	Inspect audit record	User, timestamp and quantity present	Not Executed	Not Executed
UT-SUP-01	FR-015	Supplier CRUD	Admin authenticated	Add/edit/remove supplier	Supplier record updated as requested	Not Executed	Not Executed
IT-SUP-02	FR-016	Supplier association	Registered supplier	Record stock-in	Transaction references supplier	Not Executed	Not Executed
ST-ALERT-01	FR-017	Low-stock alert	Product threshold configured	Reduce quantity below threshold	Alert appears in Staff/Admin view	Not Executed	Not Executed
ST-REPORT-01	FR-018	Date-range report	Transactions exist	Request report for range	Aggregated stock/sales report displayed	Not Executed	Not Executed
ST-REPORT-02	FR-019	Top-selling products	Sales history exists	Open report	Top-selling items ranked from recorded sales	Not Executed	Not Executed
ST-PERF-01	NFR-001	Search response time	Normal test load	Run repeated catalog searches	At least 95% complete within 2 seconds	Not Executed	Not Executed
ST-AVAIL-01	NFR-002	Availability	Running service	Observe defined business window	Availability meets 99% excluding planned	Not Executed	Not Executed

					maintenance		
ST-USE-01	NFR-003	Transaction usability	Staff test user	Complete stock-in/out	Completed in <=4 screens/steps	Not Executed	Not Executed
ST-REL-01	NFR-004	Duplicate submission	Valid stock transaction form	Double-submit/ retry	Only one transaction recorded	Not Executed	Not Executed
ST-SEC-01	SEC-001	Password storage	Test account	Inspect stored credential	Password stored as hash, not plaintext	Not Executed	Not Executed
ST-SEC-03	SEC-003	Input validation	Stock forms	Submit negative/malformed quantity	Input rejected safely	Not Executed	Not Executed
ST-SEC-05	SEC-005	Unauthenticated data access	Logged-out browser	Open protected supplier/pricing view	Access denied; no sensitive data exposed	Not Executed	Not Executed
ST-SEC-06	SEC-006	Security event logging	Test account	Generate failed login/unauthorized request	Security event logged without secrets	Not Executed	Not Executed

13.1 Requirements Traceability Matrix

Req ID	Architecture Ref	Design Ref	Test Case ID	System Test ID	Coverage
FR-001	AUTH-01	SEQ-01	UT-AUTH-01/02	ST-AUTH-01	Covered
FR-005	CATALOG-01	Product class	UT-CAT-01	ST-CAT-01	Covered
FR-010	STOCK-01	Stock service	IT-STOCK-01	ST-STOCK-01	Covered
FR-012	STOCK-02	Stock validation	UT-STOCK-02	ST-STOCK-02	Covered
FR-017	REPORT-01	SEQ-03	UT-ALERT-01	ST-ALERT-01	Covered
SEC-002	AUTH-02	RBAC middleware	ST-SEC-01	ST-AUTH-01	Covered
NFR-001	API-01	Search contract	ST-PERF-01	ST-PERF-01	Covered

14. Test Metrics & Reporting

- Planned vs executed test cases.
- Pass/fail/not-executed counts and requirement coverage.
- Defect count by severity, defect aging and reopen rate.
- Performance percentile for catalog search.
- Security test coverage for SEC-001 through SEC-006.
- Regression coverage after fixes.

Reports: execution log, defect report, performance/security evidence and final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
Team Representative	Team 8	
Faculty / Reviewer		
Final Sign-off		